

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.59_e-1)

Metric

$$ds^2 = \left(\frac{4\kappa_0 r A(x, y, u) - \kappa_0 Q_x(x, y, u)^2 - \kappa_0 Q_y(x, y, u)^2 + 4m_0 r + 2r^2}{2r^2} \right) du \otimes du - du \otimes dr - dr \otimes du \\ + \left(\frac{16r^2}{x^4 + y^4 + 2(x^2 - 4)y^2 - 8x^2 + 16} \right) dx \otimes dx + \left(\frac{16r^2}{x^4 + y^4 + 2(x^2 - 4)y^2 - 8x^2 + 16} \right) dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(-\frac{\kappa_0 A(x, y, u)}{r} + \frac{\kappa_0 Q_x(x, y, u)^2}{4r^2} + \frac{\kappa_0 Q_y(x, y, u)^2}{4r^2} - \frac{m_0}{r} - \frac{1}{2} \right) du + (1) dr$$

$$m = \left(-\frac{2\sqrt{2}r}{x^2 + y^2 - 4} \right) dx + \left(-\frac{2i\sqrt{2}r}{x^2 + y^2 - 4} \right) dy$$

$$\bar{m} = \left(-\frac{2\sqrt{2}r}{x^2 + y^2 - 4} \right) dx + \left(\frac{2i\sqrt{2}r}{x^2 + y^2 - 4} \right) dy$$

Contravariant components

$$l = (-1) \, dr$$

$$n = (-1) \, du + \left(-\frac{\kappa_0 A(x, y, u)}{r} + \frac{\kappa_0 Q_x(x, y, u)^2}{4r^2} + \frac{\kappa_0 Q_y(x, y, u)^2}{4r^2} - \frac{m_0}{r} - \frac{1}{2} \right) dr$$

$$m = \left(-\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \right) dx + \left(-\frac{i\sqrt{2}x^2}{8r} - \frac{i\sqrt{2}y^2}{8r} + \frac{i\sqrt{2}}{2r} \right) dy$$

$$\bar{m} = \left(-\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \right) dx + \left(\frac{i\sqrt{2}x^2}{8r} + \frac{i\sqrt{2}y^2}{8r} - \frac{i\sqrt{2}}{2r} \right) dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u - \frac{\kappa_0 A(x, y, u)}{r} + \frac{\kappa_0 Q_x(x, y, u)^2}{4r^2} + \frac{\kappa_0 Q_y(x, y, u)^2}{4r^2} - \frac{m_0}{r} - \frac{1}{2} \partial_r$$

$$\delta = -\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \partial_x - \frac{i\sqrt{2}x^2}{8r} - \frac{i\sqrt{2}y^2}{8r} + \frac{i\sqrt{2}}{2r} \partial_y$$

$$\bar{\delta} = -\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \partial_x + \frac{i\sqrt{2}x^2}{8r} + \frac{i\sqrt{2}y^2}{8r} - \frac{i\sqrt{2}}{2r} \partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\kappa_0 A(x, y, u)}{r^2} + \frac{\kappa_0 Q_x(x, y, u)^2}{4r^3} + \frac{\kappa_0 Q_y(x, y, u)^2}{4r^3} - \frac{m_0}{r^2} - \frac{1}{2r}$$

$$\lambda = 0$$

$$\begin{aligned} \nu = & \frac{\sqrt{2}\kappa_0 x^2 \frac{\partial}{\partial x} A(x, y, u)}{8r^2} + \frac{\sqrt{2}\kappa_0 y^2 \frac{\partial}{\partial x} A(x, y, u)}{8r^2} - \frac{i\sqrt{2}\kappa_0 x^2 \frac{\partial}{\partial y} A(x, y, u)}{8r^2} - \frac{i\sqrt{2}\kappa_0 y^2 \frac{\partial}{\partial y} A(x, y, u)}{8r^2} - \frac{\sqrt{2}\kappa_0 x^2 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{16r^3} \\ & - \frac{\sqrt{2}\kappa_0 y^2 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{16r^3} + \frac{i\sqrt{2}\kappa_0 x^2 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{16r^3} + \frac{i\sqrt{2}\kappa_0 y^2 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{16r^3} - \frac{\sqrt{2}\kappa_0 x^2 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{16r^3} \\ & - \frac{\sqrt{2}\kappa_0 y^2 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{16r^3} + \frac{i\sqrt{2}\kappa_0 x^2 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{16r^3} + \frac{i\sqrt{2}\kappa_0 y^2 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{16r^3} - \frac{\sqrt{2}\kappa_0 \frac{\partial}{\partial x} A(x, y, u)}{2r^2} + \frac{i\sqrt{2}\kappa_0 \frac{\partial}{\partial y} A(x, y, u)}{2r^2} \\ & + \frac{\sqrt{2}\kappa_0 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{4r^3} - \frac{i\sqrt{2}\kappa_0 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{4r^3} + \frac{\sqrt{2}\kappa_0 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{4r^3} - \frac{i\sqrt{2}\kappa_0 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{4r^3} \end{aligned}$$

$$\alpha = -\frac{\sqrt{2}x}{8r} + \frac{i\sqrt{2}y}{8r}$$

$$\beta = \frac{\sqrt{2}x}{8r} + \frac{i\sqrt{2}y}{8r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{\kappa_0 A(x, y, u)}{2r^2} + \frac{\kappa_0 Q_x(x, y, u)^2}{4r^3} + \frac{\kappa_0 Q_y(x, y, u)^2}{4r^3} - \frac{m_0}{2r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\kappa_0 Q_x(x, y, u)^2}{4 r^4} + \frac{\kappa_0 Q_y(x, y, u)^2}{4 r^4}$$

$$\begin{aligned} \Phi_{12} = & \frac{\sqrt{2}\kappa_0 x^2 \frac{\partial}{\partial x} A(x, y, u)}{16 r^3} + \frac{\sqrt{2}\kappa_0 y^2 \frac{\partial}{\partial x} A(x, y, u)}{16 r^3} + \frac{i \sqrt{2}\kappa_0 x^2 \frac{\partial}{\partial y} A(x, y, u)}{16 r^3} + \frac{i \sqrt{2}\kappa_0 y^2 \frac{\partial}{\partial y} A(x, y, u)}{16 r^3} - \frac{\sqrt{2}\kappa_0 x^2 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{16 r^4} \\ & - \frac{\sqrt{2}\kappa_0 y^2 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{16 r^4} - \frac{i \sqrt{2}\kappa_0 x^2 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{16 r^4} - \frac{i \sqrt{2}\kappa_0 y^2 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{16 r^4} - \frac{\sqrt{2}\kappa_0 x^2 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{16 r^4} \\ & - \frac{\sqrt{2}\kappa_0 y^2 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{16 r^4} - \frac{i \sqrt{2}\kappa_0 x^2 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{16 r^4} - \frac{i \sqrt{2}\kappa_0 y^2 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{16 r^4} - \frac{\sqrt{2}\kappa_0 \frac{\partial}{\partial x} A(x, y, u)}{4 r^3} - \frac{i \sqrt{2}\kappa_0 \frac{\partial}{\partial y} A(x, y, u)}{4 r^3} \\ & + \frac{\sqrt{2}\kappa_0 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{4 r^4} + \frac{i \sqrt{2}\kappa_0 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{4 r^4} + \frac{\sqrt{2}\kappa_0 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{4 r^4} + \frac{i \sqrt{2}\kappa_0 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{4 r^4} \end{aligned}$$

$$\Phi_{20} = 0$$

$$\begin{aligned} \Phi_{21} = & \frac{\sqrt{2}\kappa_0 x^2 \frac{\partial}{\partial x} A(x, y, u)}{16 r^3} + \frac{\sqrt{2}\kappa_0 y^2 \frac{\partial}{\partial x} A(x, y, u)}{16 r^3} - \frac{i \sqrt{2}\kappa_0 x^2 \frac{\partial}{\partial y} A(x, y, u)}{16 r^3} - \frac{i \sqrt{2}\kappa_0 y^2 \frac{\partial}{\partial y} A(x, y, u)}{16 r^3} - \frac{\sqrt{2}\kappa_0 x^2 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{16 r^4} \\ & - \frac{\sqrt{2}\kappa_0 y^2 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{16 r^4} + \frac{i \sqrt{2}\kappa_0 x^2 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{16 r^4} + \frac{i \sqrt{2}\kappa_0 y^2 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{16 r^4} - \frac{\sqrt{2}\kappa_0 x^2 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{16 r^4} \\ & - \frac{\sqrt{2}\kappa_0 y^2 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{16 r^4} + \frac{i \sqrt{2}\kappa_0 x^2 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{16 r^4} + \frac{i \sqrt{2}\kappa_0 y^2 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{16 r^4} - \frac{\sqrt{2}\kappa_0 \frac{\partial}{\partial x} A(x, y, u)}{4 r^3} + \frac{i \sqrt{2}\kappa_0 \frac{\partial}{\partial y} A(x, y, u)}{4 r^3} \\ & + \frac{\sqrt{2}\kappa_0 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{4 r^4} - \frac{i \sqrt{2}\kappa_0 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{4 r^4} + \frac{\sqrt{2}\kappa_0 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{4 r^4} - \frac{i \sqrt{2}\kappa_0 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{4 r^4} \end{aligned}$$

$$\begin{aligned}
\Phi_{22} = & -\frac{\kappa_0 x^4 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{32 r^3} - \frac{\kappa_0 x^2 y^2 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{16 r^3} - \frac{\kappa_0 y^4 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{32 r^3} - \frac{\kappa_0 x^4 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{32 r^3} - \frac{\kappa_0 x^2 y^2 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{16 r^3} - \frac{\kappa_0 y^4 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{32 r^3} \\
& + \frac{\kappa_0 x^4 \frac{\partial}{\partial x} Q_x(x, y, u)^2}{64 r^4} + \frac{\kappa_0 x^2 y^2 \frac{\partial}{\partial x} Q_x(x, y, u)^2}{32 r^4} + \frac{\kappa_0 y^4 \frac{\partial}{\partial x} Q_x(x, y, u)^2}{64 r^4} + \frac{\kappa_0 x^4 Q_x(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_x(x, y, u)}{64 r^4} + \frac{\kappa_0 x^2 y^2 Q_x(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_x(x, y, u)}{32 r^4} \\
& + \frac{\kappa_0 y^4 Q_x(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_x(x, y, u)}{64 r^4} + \frac{\kappa_0 x^4 \frac{\partial}{\partial y} Q_x(x, y, u)^2}{64 r^4} + \frac{\kappa_0 x^2 y^2 \frac{\partial}{\partial y} Q_x(x, y, u)^2}{32 r^4} + \frac{\kappa_0 y^4 \frac{\partial}{\partial y} Q_x(x, y, u)^2}{64 r^4} + \frac{\kappa_0 x^4 Q_x(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_x(x, y, u)}{64 r^4} \\
& + \frac{\kappa_0 x^2 y^2 Q_x(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_x(x, y, u)}{32 r^4} + \frac{\kappa_0 y^4 Q_x(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_x(x, y, u)}{64 r^4} + \frac{\kappa_0 x^4 \frac{\partial}{\partial x} Q_y(x, y, u)^2}{64 r^4} + \frac{\kappa_0 x^2 y^2 \frac{\partial}{\partial x} Q_y(x, y, u)^2}{32 r^4} + \frac{\kappa_0 y^4 \frac{\partial}{\partial x} Q_y(x, y, u)^2}{64 r^4} \\
& + \frac{\kappa_0 x^4 Q_y(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_y(x, y, u)}{64 r^4} + \frac{\kappa_0 x^2 y^2 Q_y(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_y(x, y, u)}{32 r^4} + \frac{\kappa_0 y^4 Q_y(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_y(x, y, u)}{64 r^4} + \frac{\kappa_0 x^4 \frac{\partial}{\partial y} Q_y(x, y, u)^2}{64 r^4} + \frac{\kappa_0 x^2 y^2 \frac{\partial}{\partial y} Q_y(x, y, u)^2}{32 r^4} \\
& + \frac{\kappa_0 y^4 \frac{\partial}{\partial y} Q_y(x, y, u)^2}{64 r^4} + \frac{\kappa_0 x^4 Q_y(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_y(x, y, u)}{64 r^4} + \frac{\kappa_0 x^2 y^2 Q_y(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_y(x, y, u)}{32 r^4} + \frac{\kappa_0 y^4 Q_y(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_y(x, y, u)}{64 r^4} + \frac{\kappa_0 x^2 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{4 r^3} \\
& + \frac{\kappa_0 y^2 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{4 r^3} + \frac{\kappa_0 x^2 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{4 r^3} + \frac{\kappa_0 y^2 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{4 r^3} - \frac{\kappa_0 x^2 \frac{\partial}{\partial x} Q_x(x, y, u)^2}{8 r^4} - \frac{\kappa_0 y^2 \frac{\partial}{\partial x} Q_x(x, y, u)^2}{8 r^4} - \frac{\kappa_0 x^2 Q_x(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_x(x, y, u)}{8 r^4} \\
& - \frac{\kappa_0 y^2 Q_x(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_x(x, y, u)}{8 r^4} - \frac{\kappa_0 x^2 \frac{\partial}{\partial y} Q_x(x, y, u)^2}{8 r^4} - \frac{\kappa_0 y^2 \frac{\partial}{\partial y} Q_x(x, y, u)^2}{8 r^4} - \frac{\kappa_0 x^2 Q_x(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_x(x, y, u)}{8 r^4} - \frac{\kappa_0 y^2 Q_x(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_x(x, y, u)}{8 r^4} \\
& - \frac{\kappa_0 x^2 \frac{\partial}{\partial x} Q_y(x, y, u)^2}{8 r^4} - \frac{\kappa_0 y^2 \frac{\partial}{\partial x} Q_y(x, y, u)^2}{8 r^4} - \frac{\kappa_0 x^2 Q_y(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_y(x, y, u)}{8 r^4} - \frac{\kappa_0 y^2 Q_y(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_y(x, y, u)}{8 r^4} - \frac{\kappa_0 x^2 \frac{\partial}{\partial y} Q_y(x, y, u)^2}{8 r^4} \\
& - \frac{\kappa_0 y^2 \frac{\partial}{\partial y} Q_y(x, y, u)^2}{8 r^4} - \frac{\kappa_0 x^2 Q_y(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_y(x, y, u)}{8 r^4} - \frac{\kappa_0 y^2 Q_y(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_y(x, y, u)}{8 r^4} - \frac{\kappa_0 \frac{\partial}{\partial u} A(x, y, u)}{r^2} + \frac{\kappa_0 Q_x(x, y, u) \frac{\partial}{\partial u} Q_x(x, y, u)}{2 r^3} \\
& + \frac{\kappa_0 Q_y(x, y, u) \frac{\partial}{\partial u} Q_y(x, y, u)}{2 r^3} - \frac{\kappa_0 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{2 r^3} - \frac{\kappa_0 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{2 r^3} + \frac{\kappa_0 \frac{\partial}{\partial x} Q_x(x, y, u)^2}{4 r^4} + \frac{\kappa_0 Q_x(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_x(x, y, u)}{4 r^4} + \frac{\kappa_0 \frac{\partial}{\partial y} Q_x(x, y, u)^2}{4 r^4} \\
& + \frac{\kappa_0 Q_x(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_x(x, y, u)}{4 r^4} + \frac{\kappa_0 \frac{\partial}{\partial x} Q_y(x, y, u)^2}{4 r^4} + \frac{\kappa_0 Q_y(x, y, u) \frac{\partial^2}{(\partial x)^2} Q_y(x, y, u)}{4 r^4} + \frac{\kappa_0 \frac{\partial}{\partial y} Q_y(x, y, u)^2}{4 r^4} + \frac{\kappa_0 Q_y(x, y, u) \frac{\partial^2}{(\partial y)^2} Q_y(x, y, u)}{4 r^4}
\end{aligned}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{\kappa_0 A(x, y, u)}{r^3} + \frac{\kappa_0 Q_x(x, y, u)^2}{2r^4} + \frac{\kappa_0 Q_y(x, y, u)^2}{2r^4} - \frac{m_0}{r^3}$$

$$\begin{aligned} \Psi_3 = & \frac{3\sqrt{2}\kappa_0 x^2 \frac{\partial}{\partial x} A(x, y, u)}{16r^3} + \frac{3\sqrt{2}\kappa_0 y^2 \frac{\partial}{\partial x} A(x, y, u)}{16r^3} - \frac{3i\sqrt{2}\kappa_0 x^2 \frac{\partial}{\partial y} A(x, y, u)}{16r^3} - \frac{3i\sqrt{2}\kappa_0 y^2 \frac{\partial}{\partial y} A(x, y, u)}{16r^3} - \frac{\sqrt{2}\kappa_0 x^2 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{8r^4} \\ & - \frac{\sqrt{2}\kappa_0 y^2 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{8r^4} + \frac{i\sqrt{2}\kappa_0 x^2 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{8r^4} + \frac{i\sqrt{2}\kappa_0 y^2 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{8r^4} - \frac{\sqrt{2}\kappa_0 x^2 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{8r^4} \\ & - \frac{\sqrt{2}\kappa_0 y^2 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{8r^4} + \frac{i\sqrt{2}\kappa_0 x^2 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{8r^4} + \frac{i\sqrt{2}\kappa_0 y^2 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{8r^4} - \frac{3\sqrt{2}\kappa_0 \frac{\partial}{\partial x} A(x, y, u)}{4r^3} + \frac{3i\sqrt{2}\kappa_0 \frac{\partial}{\partial y} A(x, y, u)}{4r^3} \\ & + \frac{\sqrt{2}\kappa_0 Q_x(x, y, u) \frac{\partial}{\partial x} Q_x(x, y, u)}{2r^4} - \frac{i\sqrt{2}\kappa_0 Q_x(x, y, u) \frac{\partial}{\partial y} Q_x(x, y, u)}{2r^4} + \frac{\sqrt{2}\kappa_0 Q_y(x, y, u) \frac{\partial}{\partial x} Q_y(x, y, u)}{2r^4} - \frac{i\sqrt{2}\kappa_0 Q_y(x, y, u) \frac{\partial}{\partial y} Q_y(x, y, u)}{2r^4} \end{aligned}$$

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$$\begin{aligned}
& -\frac{i\kappa_0x^2yQ_y(x,y,u)}{16r^4}\frac{\partial}{\partial x}Q_y(x,y,u)+\frac{\kappa_0xy^2Q_y(x,y,u)}{16r^4}\frac{\partial}{\partial x}Q_y(x,y,u)-\frac{i\kappa_0y^3Q_y(x,y,u)}{16r^4}\frac{\partial}{\partial x}Q_y(x,y,u)-\frac{i\kappa_0x^3Q_y(x,y,u)}{16r^4}\frac{\partial}{\partial y}Q_y(x,y,u) \\
& -\frac{\kappa_0x^2yQ_y(x,y,u)}{16r^4}\frac{\partial}{\partial y}Q_y(x,y,u)-\frac{i\kappa_0xy^2Q_y(x,y,u)}{16r^4}\frac{\partial}{\partial y}Q_y(x,y,u)-\frac{\kappa_0y^3Q_y(x,y,u)}{16r^4}\frac{\partial}{\partial y}Q_y(x,y,u)+\frac{\kappa_0x^2\frac{\partial^2}{(\partial x)^2}A(x,y,u)}{4r^3} \\
& +\frac{\kappa_0y^2\frac{\partial^2}{(\partial x)^2}A(x,y,u)}{4r^3}-\frac{i\kappa_0x^2\frac{\partial^2}{\partial x\partial y}A(x,y,u)}{2r^3}-\frac{i\kappa_0y^2\frac{\partial^2}{\partial x\partial y}A(x,y,u)}{2r^3}-\frac{\kappa_0x^2\frac{\partial^2}{(\partial y)^2}A(x,y,u)}{4r^3}-\frac{\kappa_0y^2\frac{\partial^2}{(\partial y)^2}A(x,y,u)}{4r^3}-\frac{\kappa_0x^2\frac{\partial}{\partial x}Q_x(x,y,u)^2}{8r^4} \\
& -\frac{\kappa_0y^2\frac{\partial}{\partial x}Q_x(x,y,u)^2}{8r^4}-\frac{\kappa_0x^2Q_x(x,y,u)}{8r^4}\frac{\partial^2}{(\partial x)^2}Q_x(x,y,u)-\frac{\kappa_0y^2Q_x(x,y,u)}{8r^4}\frac{\partial^2}{(\partial x)^2}Q_x(x,y,u)+\frac{i\kappa_0x^2Q_x(x,y,u)}{4r^4}\frac{\partial^2}{\partial x\partial y}Q_x(x,y,u) \\
& +\frac{i\kappa_0y^2Q_x(x,y,u)}{4r^4}\frac{\partial^2}{\partial x\partial y}Q_x(x,y,u)+\frac{i\kappa_0x^2\frac{\partial}{\partial x}Q_x(x,y,u)}{4r^4}\frac{\partial}{\partial y}Q_x(x,y,u)+\frac{i\kappa_0y^2\frac{\partial}{\partial x}Q_x(x,y,u)}{4r^4}\frac{\partial}{\partial y}Q_x(x,y,u)+\frac{\kappa_0x^2\frac{\partial}{\partial y}Q_x(x,y,u)^2}{8r^4} \\
& +\frac{\kappa_0y^2\frac{\partial}{\partial y}Q_x(x,y,u)^2}{8r^4}+\frac{\kappa_0x^2Q_x(x,y,u)}{8r^4}\frac{\partial^2}{(\partial y)^2}Q_x(x,y,u)+\frac{\kappa_0y^2Q_x(x,y,u)}{8r^4}\frac{\partial^2}{(\partial y)^2}Q_x(x,y,u)-\frac{\kappa_0x^2\frac{\partial}{\partial x}Q_y(x,y,u)^2}{8r^4}-\frac{\kappa_0y^2\frac{\partial}{\partial x}Q_y(x,y,u)^2}{8r^4} \\
& -\frac{\kappa_0x^2Q_y(x,y,u)}{8r^4}\frac{\partial^2}{(\partial x)^2}Q_y(x,y,u)-\frac{\kappa_0y^2Q_y(x,y,u)}{8r^4}\frac{\partial^2}{(\partial x)^2}Q_y(x,y,u)+\frac{i\kappa_0x^2Q_y(x,y,u)}{4r^4}\frac{\partial^2}{\partial x\partial y}Q_y(x,y,u)+\frac{i\kappa_0y^2Q_y(x,y,u)}{4r^4}\frac{\partial^2}{\partial x\partial y}Q_y(x,y,u) \\
& +\frac{i\kappa_0x^2\frac{\partial}{\partial x}Q_y(x,y,u)}{4r^4}\frac{\partial}{\partial y}Q_y(x,y,u)+\frac{i\kappa_0y^2\frac{\partial}{\partial x}Q_y(x,y,u)}{4r^4}\frac{\partial}{\partial y}Q_y(x,y,u)+\frac{\kappa_0x^2\frac{\partial}{\partial y}Q_y(x,y,u)^2}{8r^4}+\frac{\kappa_0y^2\frac{\partial}{\partial y}Q_y(x,y,u)^2}{8r^4} \\
& +\frac{\kappa_0x^2Q_y(x,y,u)}{8r^4}\frac{\partial^2}{(\partial y)^2}Q_y(x,y,u)+\frac{\kappa_0y^2Q_y(x,y,u)}{8r^4}\frac{\partial^2}{(\partial y)^2}Q_y(x,y,u)+\frac{\kappa_0x\frac{\partial}{\partial x}A(x,y,u)}{2r^3}-\frac{i\kappa_0y\frac{\partial}{\partial x}A(x,y,u)}{2r^3}-\frac{i\kappa_0x\frac{\partial}{\partial y}A(x,y,u)}{2r^3}-\frac{\kappa_0y\frac{\partial}{\partial y}A(x,y,u)}{2r^3} \\
& -\frac{\kappa_0xQ_x(x,y,u)}{4r^4}\frac{\partial}{\partial x}Q_x(x,y,u)+\frac{i\kappa_0yQ_x(x,y,u)}{4r^4}\frac{\partial}{\partial x}Q_x(x,y,u)+\frac{i\kappa_0xQ_x(x,y,u)}{4r^4}\frac{\partial}{\partial y}Q_x(x,y,u)+\frac{\kappa_0yQ_x(x,y,u)}{4r^4}\frac{\partial}{\partial y}Q_x(x,y,u) \\
& -\frac{\kappa_0xQ_y(x,y,u)}{4r^4}\frac{\partial}{\partial x}Q_y(x,y,u)+\frac{i\kappa_0yQ_y(x,y,u)}{4r^4}\frac{\partial}{\partial x}Q_y(x,y,u)+\frac{i\kappa_0xQ_y(x,y,u)}{4r^4}\frac{\partial}{\partial y}Q_y(x,y,u)+\frac{\kappa_0yQ_y(x,y,u)}{4r^4}\frac{\partial}{\partial y}Q_y(x,y,u) \\
& -\frac{\kappa_0\frac{\partial^2}{(\partial x)^2}A(x,y,u)}{2r^3}+\frac{i\kappa_0\frac{\partial^2}{\partial x\partial y}A(x,y,u)}{r^3}+\frac{\kappa_0\frac{\partial^2}{(\partial y)^2}A(x,y,u)}{2r^3}+\frac{\kappa_0\frac{\partial}{\partial x}Q_x(x,y,u)^2}{4r^4}+\frac{\kappa_0Q_x(x,y,u)}{4r^4}\frac{\partial^2}{(\partial x)^2}Q_x(x,y,u)-\frac{i\kappa_0Q_x(x,y,u)}{2r^4}\frac{\partial^2}{\partial x\partial y}Q_x(x,y,u) \\
& -\frac{i\kappa_0\frac{\partial}{\partial x}Q_x(x,y,u)}{2r^4}\frac{\partial}{\partial y}Q_x(x,y,u)-\frac{\kappa_0\frac{\partial}{\partial y}Q_x(x,y,u)^2}{4r^4}-\frac{\kappa_0Q_x(x,y,u)}{4r^4}\frac{\partial^2}{(\partial y)^2}Q_x(x,y,u)+\frac{\kappa_0\frac{\partial}{\partial x}Q_y(x,y,u)^2}{4r^4}+\frac{\kappa_0Q_y(x,y,u)}{4r^4}\frac{\partial^2}{(\partial x)^2}Q_y(x,y,u) \\
& -\frac{i\kappa_0Q_y(x,y,u)}{2r^4}\frac{\partial^2}{\partial x\partial y}Q_y(x,y,u)-\frac{i\kappa_0\frac{\partial}{\partial x}Q_y(x,y,u)}{2r^4}\frac{\partial}{\partial y}Q_y(x,y,u)-\frac{\kappa_0\frac{\partial}{\partial y}Q_y(x,y,u)^2}{4r^4}-\frac{\kappa_0Q_y(x,y,u)}{4r^4}\frac{\partial^2}{(\partial y)^2}Q_y(x,y,u)
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"